

Supporting Information (SI) for:

Electrostatic coupling and water bridging in adsorption hierarchy of biomolecules at water-clay interfaces

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This document contains:

Materials and Methods

SI Appendix Figs. S1-S11

SI Appendix Tables S1-S6

MATERIALS AND METHODS

1. Chemicals. A Na-rich montmorillonite (MMT, Swy-3), with average stoichiometry of $[\text{Ca}_{0.12}\text{Na}_{0.32}\text{K}_{0.05}][\text{Al}_{3.01}\text{Fe(III)}_{0.41}\text{Mn}_{0.01}\text{Mg}_{0.54}\text{Ti}_{0.02}][\text{Si}_{7.98}\text{Al}_{0.02}]\text{O}_{20}(\text{OH})_4$, was purchased from the Clay Mineral Society (West Lafayette, IN, U.S.A). Ultra-purified water was generated onsite (18.2 MΩ cm) and used for all solutions. Xylose (purity > 98%) and *p*-coumaric acid (purity > 98%) were purchased from MP Biomedicals and Tokyo Chemical Industry Co., respectively. The rest of chemicals (purity > 98%) including lysine, histidine, glutamine, leucine, phenylalanine, threonine, glucose, glutamate, sodium bicarbonate, sodium chloride, and magnesium chloride used in this research were purchased from Sigma-Aldrich.

2. Adsorption experiments. We conducted adsorption experiments with solutions containing either a single compound or a mixture of all ten biomolecules with different types of organic functional groups and structures. The background solution contained 50 mM NaHCO₃ and 100 mM NaCl (or 33.3 mM MgCl₂ for constant ionic strength) with pH was adjusted to 7.0 ± 0.05 using drops of 6 N HCl. For individual adsorption experiments, stock solution (4 mM) of each compound was prepared in the background solution and diluted to different concentrations (125, 250, 500, 1000, 2000, 4000 μM). We mixed 20 mL of the prepared solution with 1 g of MMT in 50-mL polypropylene tubes (Corning Inc, Corning, NY) before the incubation in an incubator-shaker (Eppendorf, Hamburg, Germany) under dark conditions at 25 °C and 150 rpm for 24 h, which was demonstrated to be long enough to reach the adsorption equilibrium for all investigated compounds in preliminary experiments. Blank experiments were also performed in the absence of clay, and three replicates were prepared for all samples. For adsorption experiments with a mixture of compounds, we followed the aforementioned procedure but dissolved all ten compounds together with the same concentration gradient as previously mentioned to the background solution before the adding of MMT. Informed by obtained adsorption results, we attempted to identify specifically the pairs of compounds competing with each other using following combinations of compounds for adsorption: lysine and histidine, lysine and threonine, lysine and glutamate, lysine and *p*-coumarate. After incubation, samples were centrifuged at 4000 rpm (3739 ×*g*) for 30 mins to separate MMT from the mixture. The

supernatant was filtered using 0.22- μm membranes (Agilent, Santa Clara, CA) and the concentration of each compound was determined via ^1H nuclear magnetic resonance (NMR) for following analysis.

The equilibrium adsorption amount of each organic compound on MMT (Q , μmol compound g^{-1} clay) was calculated using equation (1) where m is the concentration of MMT (50 g L^{-1}), C_{eq} ($\mu\text{mol L}^{-1}$) is the concentration of each organic in solution at equilibrium, and C_0 ($\mu\text{mol L}^{-1}$) is the concentration of corresponding blank control without MMT. The adsorption isotherm graph, which was prepared by plotting Q as a function of C_{eq} , was fitted by equation (2) wherein K_d is the adsorption distribution coefficient for each compound.

$$Q = (C_0 - C_{eq})/m \quad (1)$$

$$Q = K_d \times C_{eq} \quad (2)$$

3. NMR quantification of compounds in solution. The NMR measurements were conducted at the Integrated Molecular Structure Education and Research Center (IMSERC) at Northwestern University (Evanston, IL) using a Bruker NOE 600 MHz spectrometer equipped with QCI-F cryoprobe for water suppression ^1H -NMR experiments. To prepare each sample for ^1H NMR analysis, we mixed 485 μL of standard solution or filtered supernatant from the adsorption experiments with 10 μL of 27.5 mM sodium trimethylsilylpropanesulfonate (DSS; Sigma-Aldrich, St. Louis, MO), and 55 μL of D_2O (Sigma-Aldrich, St. Louis, MO) in an NMR tube with 5-mm diameter (Norell, Morganton, NC). Preliminary NMR data were obtained for each biomolecule individually to identify the characteristic NMR peaks. Subsequently, a standard mixture with all ten biomolecules was prepared and analyzed by the same method to ensure the selected characteristic peaks from the different biomolecules were well-resolved with no overlaps. Finally, a calibration curve was prepared for each compound by measuring standard samples with known biomolecule concentration. This calibration curve was used to quantify the concentration of biomolecules during the adsorption experiments. The ^1H peak of the methyl group in DSS was used as a chemical shift reference to calibrate the peaks for the other compounds. The number of scans was 32 and the relaxation delay was set to 5 s. Quality control samples with known concentrations were inserted into the measurement queues after

every six samples. Baseline correction and phase correction of measured data were done using the MestReNova 14.3.1 software (Mestrelab Research, S.L, Spain).

4. X-ray diffraction characterization of clay interlayer adsorption. First, we conducted batch adsorption experiments as mentioned in section 2 with following concentration of biomolecules: 150, 300, 600, 1500, 3000, and 6000 μM . After the removal of the supernatant from adsorption samples, the clay pellets were air-dried and analyzed using a Bruker D8 Advance powder X-ray diffractometer. To maintain constant temperature (25 °C) and relative humidity (20%) throughout the X-ray diffraction measurements, samples were placed in a CHC+ humidity-controlled chamber (Anton Paar, Graz, Austria) attached with an MHG-32 modular humidity generator (ProUmid, Ulm, Germany) and a TCU110 temperature control unit (Anton Paar, Graz, Austria). Data were collected with a step size of 0.02° and 8 s per step over the 2–10° 2 θ Cu K α angular range (λ = 1.5418 Å). The Diffrac.suite software from Bruker was used to process measured X-ray diffraction data and calculate the interlayer distance (d_{001}) of MMT.

5. Molecular Simulations. The Materials Studio software packages were used for molecular simulations in this study [1]. Chemical structures of the ten model biomolecules (used in the adsorption experiments described above) were downloaded from PubChem. Partial charges of glutamine, glutamate, and threonine were assigned by reconstructing the electrostatic potentials obtained from density-functional-theory-based geometry optimization using B3LYP function with transition state correction and DNP basis sets in Dmol3 module [2]. For the other biomolecules, partial charges were assigned using chemistry at Harvard macromolecular mechanics (CHARMM) force field [3]. We used the condensed-phase optimized molecular potentials for atomistic simulation studies (COMPASS, version II) force field with the Ewald method for electrostatic interactions and atom-based method for Van der Waals interactions [4]. Using a published protocol to simulate the biomolecules with geometry optimization and energy minimization [5], we confirmed the suitable applicability of COMPASS for simulations of these biomolecules by obtaining greater than 80% agreement between our simulated bond lengths and bond angles and those reported data from experiments [6-14]. Optimized structures of the

biomolecules were complexed with Na^+ or Mg^{2+} to be used as starting adsorptives for interactions in the hydrated clay systems. The clay used is a Wyoming-type MMT that was prepared previously [15]. The MMT unit cell with an average stoichiometry of $(\text{Si}_8)(\text{Al}_{3.5}\text{Mg}_{0.5})\text{O}_{20}(\text{OH})_4$ was extended to obtain a supercell ($42.24 \times 36.56 \times 45.00 \text{ \AA}$) with a net charge of -16, which is balanced by either Na^+ or Mg^{2+} .

To investigate the interaction mechanisms, thermodynamically-optimized initial configurations of the organo-mineral complexes were first obtained using a Monte Carlo search implemented in the Adsorption Locator module of Material Studio [16]. Two Na^+ or MgCl^+ were randomly removed from the charge-balanced MMT surface, after which two organic adsorptives, three NaCl (or MgCl_2), and SPC/E water molecules (1 g cm^{-3}) were imported. The intermolecular interactions were simulated with the same methods as aforementioned. The loaded adsorptives including water molecules were subjected to translation and rotation relative to the basal surface of MMT during an annealing process consisting of 10 temperature cycles ranging from 298 K-500 K. To obtain the thermodynamically equilibrated adsorbate structures, the configuration from the Monte Carlo search was subjected to an extra 10-ns molecular dynamics (MD) simulation using NVT ensemble and Berendsen thermostat with randomly assigned initial velocity for each molecule and constant average temperature at 298 K. One frame of configuration will be generated for every 5-ps simulation with a step size of 1 fs. For each frame in the results of MD simulations, we developed a Perl script to monitor the formation of hydrogen bond (with a maximum allowable distance of $2.5 \text{ \AA}$ and a minimum allowable angle of 120° between H-donor and H-acceptor) and cation complexation (based on Van der Waals radius) around the biomolecules, by which the different interaction mechanisms such as hydrogen bonding, water bridging, and cation bridging can be identified and quantified. Accordingly, inner sphere, outer sphere, diffusion swarm, and bulk solution can be distinguished. In the localized systems with biomolecules, bridging water, and MMT, we calculated the energy of electrostatic and Van der Waals interaction.

To capture the interactions between lysine, histidine, and *p*-coumarate during adsorption, a 50-ns MD simulation was conducted. The biomolecules were positioned initially in the center of two layers of MMT with d_{001} of $60 \text{ \AA}$ before the loading of NaCl (or MgCl_2) and SPC/E water

molecules, which resulted in a charge-balanced system with a water density of 1 g cm^{-3} . The same setup of MD simulation as last section was used. And a similar Perl script was developed to monitor the coupling of lysine and *p*-coumarate via hydrogen bonds and calculate the organic-organic or organic-clay interaction energy in each frame.

6. Statistical analysis. Linear fitting function based on least square method was used to fit adsorption isotherm and calculate K_d . Significance was determined by unpaired two-tailed Student's t-test.

SI Appendix

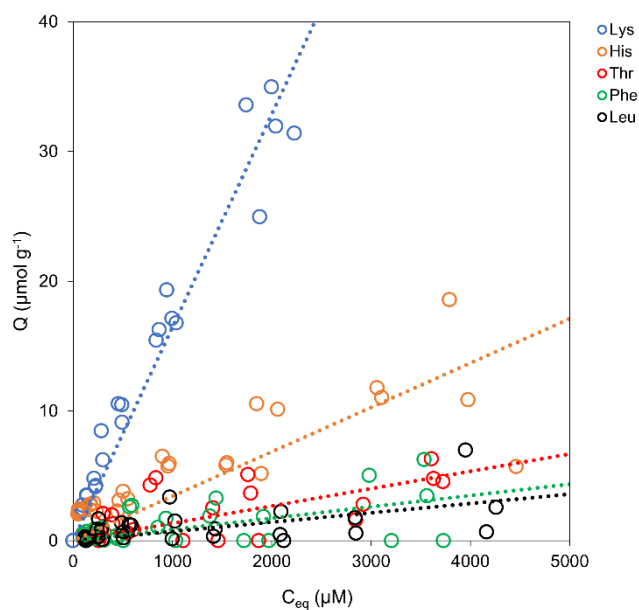


Figure S1. Adsorption isotherm of lysine, histidine, threonine, phenylalanine, and leucine, as single adsorptive, on Na-MMT. Dashed lines were obtained from linear fitting.

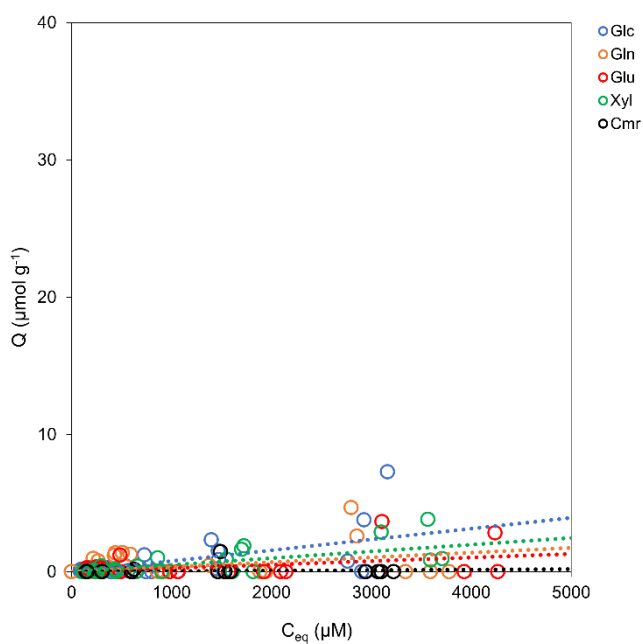


Figure S2. Adsorption isotherm of glucose, glutamine, glutamate, xylose, and *p*-coumarate, as single adsorptive, on Na-MMT. Dashed lines were obtained from linear fitting.

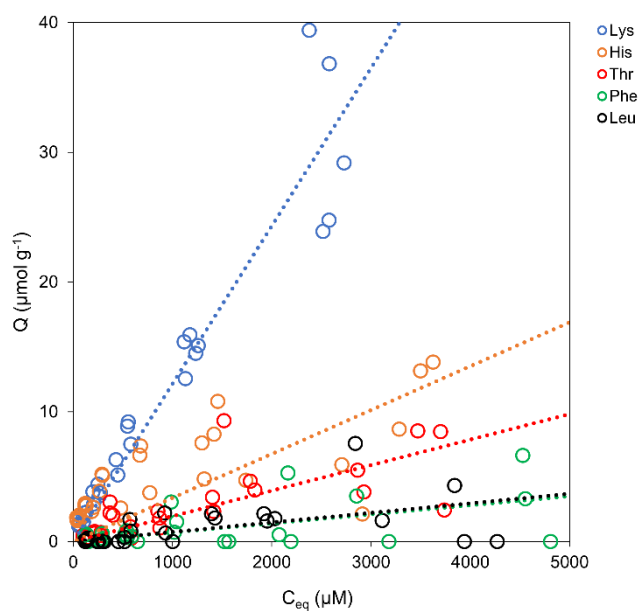


Figure S3. Adsorption isotherm of lysine, histidine, threonine, phenylalanine, and leucine, as single adsorptive, on Mg-MMT. Dashed lines were obtained from linear fitting.

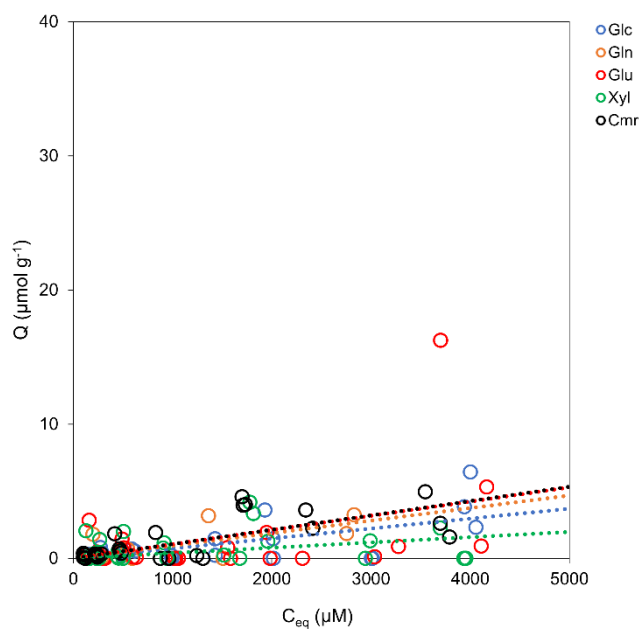


Figure S4. Adsorption isotherm of glucose, glutamine, glutamate, xylose, and *p*-coumarate, as single adsorptive, on Mg-MMT. Dashed lines were obtained from linear fitting.

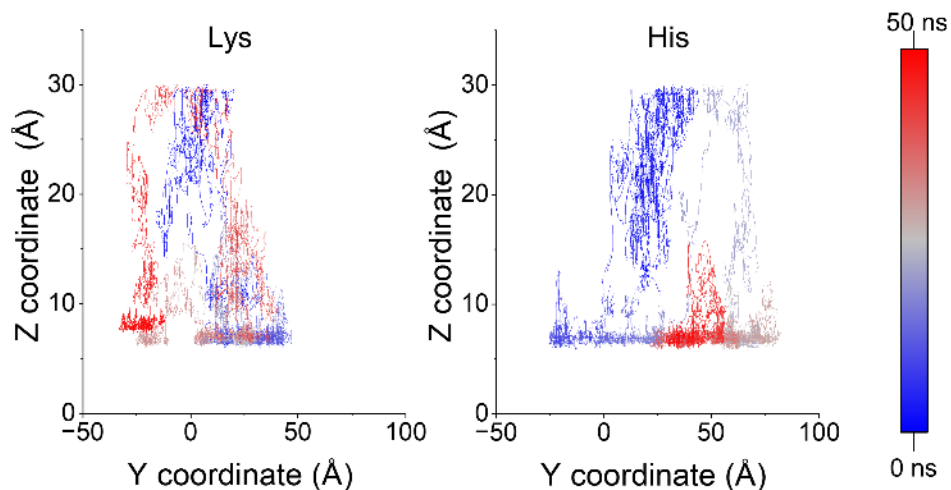


Figure S5. Diffusion trajectory of lysine (left) and histidine (right) in Na-MMT system over 50-ns MD simulations.

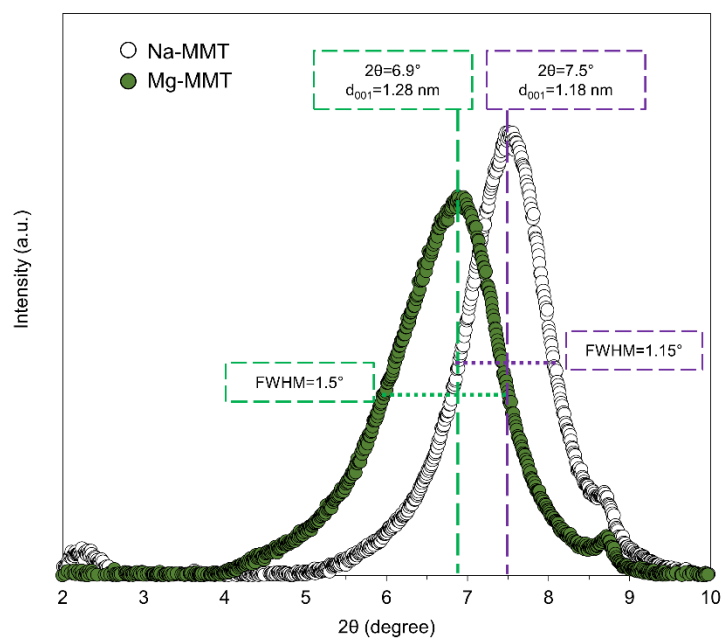


Figure S6. X-ray diffraction patterns of Na-MMT (white) and Mg-MMT (green) without organic biomolecules. The interlayer distance (d_{001}) and full width at half maximum (FWHM) were calculated and displayed for each system.

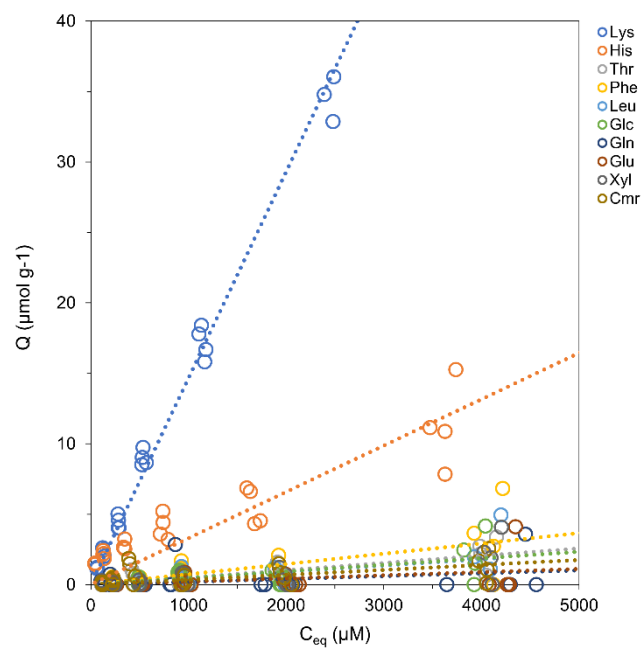


Figure S7. Adsorption isotherm of ten biomolecules, in a mixed-compound solution, on Na-MMT. Dashed lines were obtained from linear fitting.

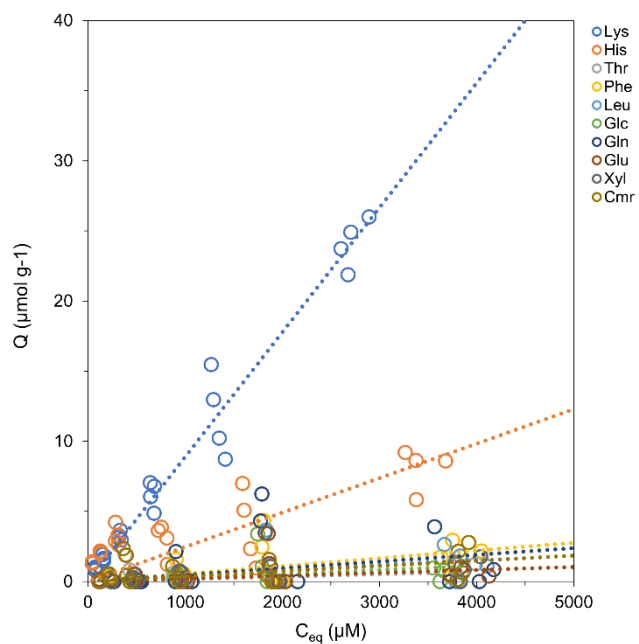


Figure S8. Adsorption isotherm of ten biomolecules, in a mixed-compound solution, on Mg-MMT. Dashed lines were obtained from linear fitting.

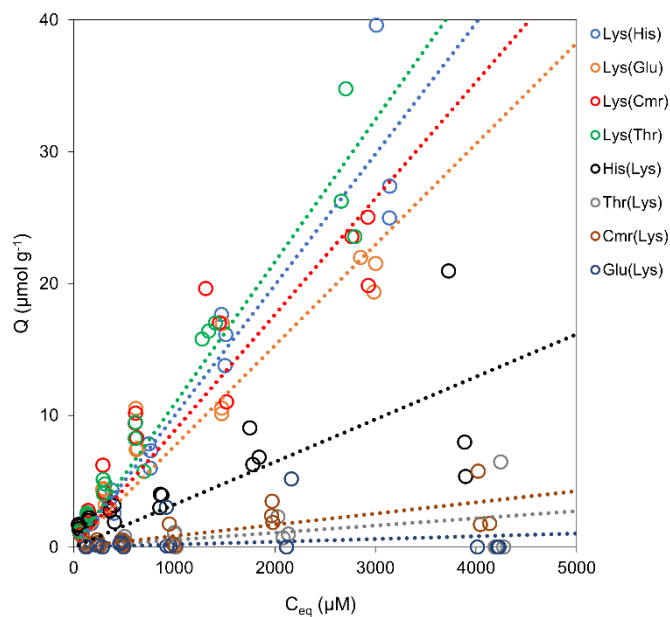


Figure S9. Adsorption isotherm of biomolecules with the presence of another organic biomolecule. For instance, Lys (His) indicated the adsorption isotherm of lysine with the presence of histidine.

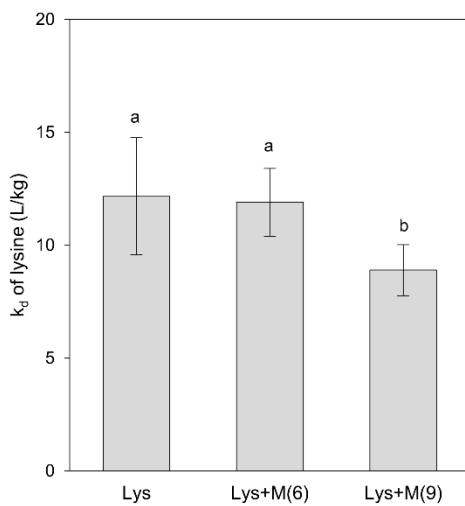


Figure S10. Adsorption distribution coefficient (K_d) of lysine on Mg-MMT with the presence of different biomolecules. M (6) is the mixture of six biomolecules including glucose, glutamine, leucine, phenylalanine, threonine, and xylose. M (9) is the mixture of nine biomolecules including *p*-coumarate, glucose, glutamate, glutamine, histidine, leucine, phenylalanine, threonine, and xylose.

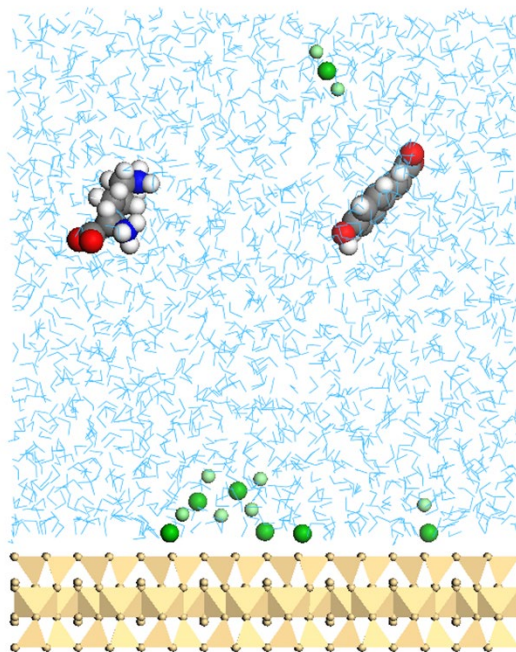


Figure S11. Initial configurations of lysine and *p*-coumarate in fully hydrated Mg-MMT system before 50-ns MD simulations; the biomolecules were placed in the bulk water, ~2.5 nm from the MMT surface. Color scheme: MMT (light yellow), water (blue), hydrogen (white), oxygen (red), nitrogen (dark blue), sodium (purple), chloride (light green).

Table S1. Adsorption distribution coefficient (K_d) of biomolecules on Na-MMT and Mg-MMT.

	K_d of replicate (L/kg)		K_d of all data (L/kg)		Standard Deviation (L/kg)	
	Na-MMT	Mg-MMT	Na-MMT	Mg-MMT	Na-MMT	Mg-MMT
Lys	15.25	11.12	16.43	12.16	2.64	2.60
	16.27	15.57				
	17.47	13.76				
	15.08	8.57				
	14.25	10.95				
His	5.05	4.59	3.42	3.38	1.66	2.06
	3.42	3.70				
	1.71	3.45				
	1.53	2.11				
	2.14	1.31				
Thr	1.71	2.33	1.34	1.97	0.92	1.10
	1.18	3.02				
	1.78	1.07				
	0.17	1.72				
	0.25	1.49				
Phe	1.36	1.69	0.87	0.70	0.88	0.75
	0.26	0.03				
	0.74	0.68				
	3.75	2.61				
	2.10	0.01				
Leu	0.50	0.21	0.72	0.74	0.71	0.84
	1.53	0.27				
	0.42	0.99				
	1.11	0.18				
	1.87	2.42				
Glc	1.17	1.36	0.78	0.88	0.74	0.63
	1.97	1.08				
	0.00	0.45				
	0.52	2.72				
	0.01	0.62				
Gln	0.00	3.64	0.35	0.94	0.66	0.71
	0.05	4.72				
	0.04	2.16				
	0.67	0.30				
	0.20	0.22				
Glu	0.50	0.95	0.26	1.06	0.43	1.54
	0.00	3.40				
	0.03	0.19				
	0.00	2.39				
	0.30	0.33				
Xyl	0.35	0.13	0.49	0.39	0.44	0.68
	1.06	0.41				
	0.20	0.85				
	0.17	0.03				
	0.03	0.17				
Cmr	0.00	0.71	0.04	1.07	0.12	0.74
	0.00	0.98				
	0.00	1.66				
	0.15	0.68				
	0.00	0.42				

Table S2. Interlayer distance of MMT with different loading of biomolecules

	Lys		His		Thr		Phe		Cmr		Glu	
	Q	d ₀₀₁	Q	d ₀₀₁	Q	d ₀₀₁	Q	d ₀₀₁	Q	d ₀₀₁	Q	d ₀₀₁
Na-MMT	53.8	1.242	5.6	1.207	0.0	1.206	28.2	1.195	0.0	1.193	0.0	1.19
	33.6	1.224	11.9	1.213	0.5	1.212	13.1	1.219	0.0	1.204	0.7	1.185
	25.0	1.195	11.0	1.204	2.8	1.191	0.0	1.198	0.0	1.217	0.0	1.215
	16.3	1.193	11.8	1.197	1.6	1.187	5.0	1.212	0.0	1.198	3.7	1.198
	15.5	1.197	5.8	1.12	0.0	1.179	1.9	1.198	0.0	1.208	0.0	1.186
	8.5	1.198	6.0	1.195	2.5	1.185	3.3	1.197	0.0	1.184	0.0	1.191
	6.2	1.186	3.2	1.192	0.6	1.191	2.7	1.192	0.0	1.196	0.0	1.173
	3.5	1.186	3.8	1.194	0.8	1.191	2.6	1.192	0.0	1.198	0.0	1.195
	3.5	1.199	2.9	1.192	0.7	1.181	0.1	1.197	0.0	1.185	0.3	1.185
	2.1	1.178	2.3	1.183	0.4	1.118	1.0	1.187	0.0	1.185	0.3	1.183
	2.1	1.19	2.2	1.187	0.5	1.188	0.7	1.184	0.0	1.184	0.1	1.17
Mg-MMT					0.3	1.178	0.3	1.187	0.0	1.189	0.3	1.208
	41.5	1.275	13.0	1.284	9.9	1.278	18.5	1.263	2.8	1.252	19.1	1.266
	54.3	1.273	6.0	1.29	9.2	1.284	0.0	1.249	1.6	1.27	2.4	1.267
	24.8	1.276	2.1	1.292	5.5	1.262	3.5	1.201	3.6	1.28	0.9	1.216
	23.9	1.266	5.9	1.278	3.8	1.252	0.0	1.238	2.2	1.256	0.1	1.258
	15.4	1.285	7.6	1.272	3.4	1.247	0.0	1.232	0.0	1.266	0.0	1.226
	15.9	1.262	4.8	1.284	2.3	1.257	0.0	1.232	0.2	1.286	0.8	1.24
	5.1	1.279	2.6	1.282	1.1	1.277	0.0	1.202	0.5	1.262	0.1	1.231
	6.3	1.267	1.6	1.276	0.2	1.251	0.4	1.205	0.4	1.213	0.1	1.213
	3.8	1.284	2.6	1.272	0.7	1.275	0.2	1.215	0.1	1.282	0.0	1.234
	2.9	1.27	2.3	1.276	0.5	1.26	0.6	1.217	0.2	1.199	2.8	1.241
	1.9	1.266	2.0	1.26	0.3	1.267	0.6	1.201	0.0	1.252	0.0	1.276
	1.5	1.291	1.7	1.263	0.1	1.244	0.0	1.243	0.0	1.249	0.0	0

* Q: organic loading of biomolecules, with unit of $\mu\text{mol/g}$.* d₀₀₁: interlayer distance of MMT, with unit of nm.

Table S3. Biomolecule-clay interaction energy from molecular simulations

		Median	1st Quartile	3rd Quartile	Max	Min
Electrostatic Interaction Energy (kcal/mol)	Lys	-249.3	-211.7	-282.9	8.0	-398.7
	His	-152.0	-98.5	-184.7	-29.3	-293.1
	Thr	-4.2	36.9	-31.8	80.0	-121.0
	Phe	-127.2	-71.1	-161.0	22.0	-248.1
	Glc	2.4	23.1	-19.9	92.6	-104.9
	Leu	-112.8	-74.4	-156.7	8.6	-271.4
	Xyl	12.3	35.0	-9.8	116.4	-102.1
	Gln	-127.4	-101.7	-151.5	-22.6	-254.8
	Glu	-25.0	-8.2	-37.7	77.6	-153.7
	Cmr	-74.1	-52.3	-96.2	82.3	-216.2
Van der Waals Interaction Energy (kcal/mol)	Lys	11.2	17.3	5.6	87.6	-6.6
	His	2.2	7.4	-0.2	179.6	-10.5
	Thr	5.9	10.5	2.5	73.4	-6.1
	Phe	4.9	9.1	1.4	75.7	-7.5
	Glc	9.9	15.0	5.5	66.0	-10.2
	Leu	3.8	9.0	2.0	70.3	-7.5
	Xyl	7.4	12.1	3.7	111.8	-5.0
	Gln	5.5	10.3	2.1	97.4	-10.1
	Glu	8.2	15.7	5.6	1611.1	-4.1
	Cmr	7.3	12.0	3.8	243.0	-8.7

Table S4. Adsorption distribution coefficient (K_d) of biomolecules on Na-MMT and Mg-MMT in mixture solution with all ten compounds

	Na		Mg	
	K_d (L/kg)	s.t.d (L/kg)	K_d (L/kg)	s.t.d (L/kg)
Phe	0.73	0.61	0.55	0.54
His	3.29	1.10	2.46	1.22
cmr	0.35	0.35	0.37	0.46
Glc	0.47	0.39	0.27	0.39
Xyl	0.46	0.36	0.32	0.39
Lys	14.65	1.06	8.89	1.13
Gln	0.20	0.45	0.48	0.88
Glu	0.23	0.39	0.21	0.39
Thr	0.52	0.32	0.17	0.30
Leu	0.45	0.44	0.39	0.44

*s.t.d: standard deviation

Table S5. Adsorption of biomolecules with the presence of another biomolecule. For instance, Lys(His) indicates the K_d of lysine with the presence of histidine.

	K_d (L/Kg)	s.t.d (L/kg)
Lys(His)	9.9	2.1
Lys(Glu)	7.7	1.8
Lys(Cmr)	8.8	2.5
Lys(Thr)	10.8	2.0
His(Lys)	3.2	1.9
Glu(Lys)	0.2	1.1
Cmr(Lys)	0.9	1.1
Thr(Lys)	0.5	0.7

*s.t.d: standard deviation

Table S6. Biomolecule-biomolecule and biomolecule-clay electrostatic interaction energy

	Median	1st Quartile	3rd Quartile	Max	Min
Lys-Cmr	-218.7	-225.8	-208.5	-109.5	-254.1
Lys-MMT	-178.5	-191.5	-162.9	-65.1	-241.5
Cmr-MMT	167.3	159.1	174.1	232.5	90.9

*unit kcal/mol

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